

# Babyland Patient App – Knowledge Transfer Document

**Last updated:** March 5, 2026

**Platform:** Flutter (Dart)

**Min SDK:** ^3.9.2

---

## Table of Contents

1. [Project Overview](#)
  2. [Tech Stack & Dependencies](#)
  3. [Project Structure](#)
  4. [Entry Point & App Bootstrap](#)
  5. [State Management](#)
  6. [Navigation & Routing](#)
  7. [Data Layer \(Network / API\)](#)
  8. [Local Storage](#)
  9. [Authentication](#)
  10. [Feature Breakdown](#)
  11. [Theme & Styling](#)
  12. [Reusable Widgets](#)
  13. [Environment Configuration](#)
  14. [Existing Documentation](#)
  15. [Code Conventions](#)
  16. [Known Gotchas & Notes](#)
  17. [How to Run](#)
- 

## Project Overview

**Babyland** is a maternal & child health Flutter app that covers three life stages:

| Stage         | Description                                                       |
|---------------|-------------------------------------------------------------------|
| Pre-Pregnancy | Menstrual cycle tracking, daily logs, AI insights, cycle calendar |

|                       |                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------|
| <b>Pregnancy</b>      | Pregnancy dashboard, fetal development, appointments, save logs, community                    |
| <b>Post-Pregnancy</b> | Baby details, baby growth (milestones, photos, vaccinations), recovery tracking, feeding logs |

The app also includes **Expert Consultation** (doctor listing, slot booking, video calling via Agora), an **AI Assistant** (chat-based), **Subscriptions**, and **User Profile** management.

---

## Tech Stack & Dependencies

| Category                | Package(s)                                                                                           |
|-------------------------|------------------------------------------------------------------------------------------------------|
| <b>State Management</b> | <code>provider</code> (ChangeNotifier + MultiProvider)                                               |
| <b>Networking</b>       | <code>dio</code> (HTTP client)                                                                       |
| <b>Auth</b>             | <code>firebase_auth</code> , <code>google_sign_in</code> , <code>googleapis_auth</code>              |
| <b>Firestore</b>        | <code>firebase_core</code> , <code>firebase_messaging</code>                                         |
| <b>Local Storage</b>    | <code>flutter_secure_storage</code> (tokens, user data), <code>hive_flutter</code> (first-time flag) |

|                      |                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Video Calling</b> | <code>agora_rtc_engine</code>                                                                                                    |
| <b>AI Chat / STT</b> | <code>speech_to_text</code>                                                                                                      |
| <b>UI</b>            | <code>cached_network_image, flutter_svg, shimmer, table_calendar, pinput, flutter_html, flutter_spinkit, dropdown_button2</code> |
| <b>Utilities</b>     | <code>intl, image_picker, file_picker, url_launcher, permission_handler, flutter_dotenv</code>                                   |
| <b>Env</b>           | <code>flutter_dotenv</code> – loads from <code>assets/.env</code>                                                                |

**No new dependencies** should be added without explicit approval.

## Project Structure

```

babyland/
├── assets/
│   ├── .env                # Environment variables (SERVERClientID)
│   ├── fonts/              # Gilroy font family (10 weights)
│   └── images/              # All image assets (63 files)
├── docs/
│   ├── API_DOCUMENTATION.md # Full API docs
│   ├── CURL_EXAMPLES.md    # cURL examples for every endpoint
│   └── swagger.json        # OpenAPI spec
```

```

├─ lib/

|   ├─ main.dart                # Entry point, MultiProvider setup
|   ├─ firebase_options.dart    # Firebase config (auto-generated)
|   └─ app/

|       ├─ agora/                # Video call (Agora RTC)
|       ├─ common_model/         # Shared data models
|       ├─ common_profile_header/ # GetUserProvider, GetUserModel,
ProfileHeader widget
|       ├─ constants/           # Enums (FlowType) and image path
constants
|       ├─ controller/          # All ChangeNotifier controllers (14
subdirectories)
|       ├─ data/                # Network, repository, response, storage
|       ├─ navbar/              # Bottom navigation for pregnancy &
post-pregnancy flows
|       ├─ push_notification/    # Firebase push notification setup
|       ├─ routes/              # app_routes.dart - all named routes
|       ├─ services/            # Social login, user preferences
|       ├─ theme/               # Colors, font families, text styles
|       ├─ view/                # All screens (17 subdirectories)
|       └─ widgets/             # 28 reusable widget files

├─ pubspec.yaml
├─ structure.md                 # Existing brief structure doc
└─ firebase.json               # Firebase hosting config

```

---

## Entry Point & App Bootstrap

File: `lib/main.dart`

### Boot sequence:

1. `WidgetsFlutterBinding.ensureInitialized()`
2. `UserPreference.init()` → Initializes Hive storage
3. `Firebase.initializeApp()` → Initializes Firebase with platform options
4. `dotenv.load(fileName: "assets/.env")` → Loads environment variables
5. Gets Firebase Messaging token (for push notifications)
6. `runApp(Babyland())`

### Global instances:

```
final navigatorKey = GlobalKey<NavigatorState>(); // Global navigator key
```

```
final networkApi = NetworkApiServices(); // Dio singleton
```

```
final repository = Repository(apiService: networkApi); // Repository singleton
```

### App config:

- Portrait-only orientation
- Transparent status bar
- `initialRoute` → `AppRoutes.splashView`
- Route generation → `AppRoutes.generateRoute`

---

## State Management

Pattern: `Provider` + `ChangeNotifier`

Global Providers (injected in `main.dart` via `MultiProvider`):

| Provider                    | Purpose                                      |
|-----------------------------|----------------------------------------------|
| Repository                  | API calls (injected as <code>.value</code> ) |
| NavBarProvider              | Pregnancy flow bottom navigation             |
| ExpertConsultationProvider  | Doctor listing / filtering                   |
| BookingController           | Appointment booking logic                    |
| PostPregnancyNavBarProvider | Post-pregnancy flow bottom navigation        |
| PhotoProvider               | Baby photo management                        |
| PregnancyController         | Pregnancy data & dashboard                   |
| PostpregnancyProvider       | Post-pregnancy data                          |
| CycleCalenderProvider       | Menstrual cycle calendar                     |
| ForgotPassController        | Forgot password flow                         |
| GetUserProvider             | Current user profile fetch                   |
| BabyGrowthProvider          | Baby growth tracking                         |

|                                         |                                      |
|-----------------------------------------|--------------------------------------|
| <code>PrePregnancyNavBarProvider</code> | Pre-pregnancy flow bottom navigation |
| <code>AiAssistantProvider</code>        | AI chat room management              |
| <code>SubscriptionProvider</code>       | Subscription plans                   |
| <code>PoliciesProvider</code>           | Privacy, terms, refund policies      |
| <code>RecordsProvider</code>            | Medical records upload/view          |
| <code>VideoCallProvider</code>          | Agora video call state               |
| <code>PostAppointmentController</code>  | Post-pregnancy appointments          |

### How to access state:

```
// Read & rebuild on change:
```

```
Consumer<MyController>(builder: (_, ctrl, __) => Text(ctrl.value));
```

```
// Trigger actions without rebuild:
```

```
Provider.of<MyController>(context, listen: false).doSomething();
```

**Rule:** Do NOT change state management patterns. Stick to Provider/ChangeNotifier.

---

## Navigation & Routing

**File:** `lib/app/routes/app_routes.dart`

- **58 named routes** defined as `static const String` fields

- Route generation via `generateRoute(RouteSettings settings)` → `switch-case` mapping
- All routes use `MaterialPageRoute`
- Some routes pass `settings` (for arguments): `stagesView`, `processingDetailsView`, `signInView`, `navableView`, `dailyLogs`, `doctorProfileView`, `selectSlotTimeView`, `bookingsView`, `saveLogView`, `profileScreen`

## Key navigation patterns:

// Push named route:

```
Navigator.pushNamed(context, AppRoutes.homeView);
```

// Push & clear entire stack (e.g., after login):

```
Navigator.pushNamedAndRemoveUntil(context, AppRoutes.signInView, (route)
=> false);
```

// Pass arguments:

```
Navigator.pushNamed(context, AppRoutes.stagesView, arguments:
{"fromLoginScreen": true});
```

## Route groups:

| Group      | Routes                                                                                                                            |
|------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Onboarding | <div data-bbox="305 1522 1039 1558"> <code>splashView</code>, <code>onboardingView</code>, <code>letsGetStartedView</code> </div> |



|                            |                                                                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Auth</b>                | signInView, createNewAccountView, addEmailView, accountCreatedSuccessfullyView, forgotPasswordView, forgotPasswordOtpVerifyView, createNewPasswordView, allSetView |
| <b>Setup</b>               | basicInfoView, stagesView, processingDetailsView                                                                                                                   |
| <b>Pre-Pregnancy</b>       | navbarPrePregnancyView, menstrualCycle, dailyLogs, cycleCalendarView, aiInsights                                                                                   |
| <b>Pregnancy</b>           | navbarView, pregnancyView, pregnancyHomeView, fetalDevelopmentView, appointmentView, addReminderView, saveLogView                                                  |
| <b>Post-Pregnancy</b>      | postPregnancyNavbarView, postPregnancyView, babyDetailView, combinedBabyDetailScreen, postPreBabyGrowthView, postAppointmentView, postAddReminderView              |
| <b>Baby Growth</b>         | babyGrowthView, babyGrowthSummaryView, mildStonesView, addMildStones, photoView, addNewPhotoView, vaccinationView                                                  |
| <b>Expert Consultation</b> | allDoctorView, doctorProfileView, selectSlotTimeView, bookingsView, myBookingsView, recordsView, reportView                                                        |

|                     |                                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>AI</b>           | <code>aiAssistantView</code>                                                                                                   |
| <b>Subscription</b> | <code>prePreSubscriptionView, preSubscriptionView, postPreSubscriptionView, babyChildSubscriptionView</code>                   |
| <b>Profile</b>      | <code>profileScreen, profileUpdateScreen, privacyPolicy, termOfServicesScreen, refundPolicyScreen, shippingPolicyScreen</code> |

## Data Layer (Network / API)

### Architecture

Controller → Repository → NetworkApiServices (Dio) → Backend API

### Files:

| File                                   | Location                              | Purpose                                   |
|----------------------------------------|---------------------------------------|-------------------------------------------|
| <code>end_points.dart</code>           | <code>lib/app/data/network/</code>    | All API endpoint URLs as static constants |
| <code>network_api_services.dart</code> | <code>lib/app/data/network/</code>    | Dio singleton with interceptors           |
| <code>repository.dart</code>           | <code>lib/app/data/repository/</code> | 65+ methods mapping features to API calls |

|                                |                                      |                                                                                                                                  |
|--------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <code>api_response.dart</code> | <code>lib/app/data/responses/</code> | Generic <code>ApiResponse&lt;T&gt;</code> wrapper with <code>LOADING</code> , <code>COMPLETED</code> , <code>ERROR</code> status |
| <code>status.dart</code>       | <code>lib/app/data/responses/</code> | <code>ApiStatus</code> enum                                                                                                      |

## Base URL

`https://api-babyland.duckdns.org/api`

## Network Layer (`NetworkApiServices`)

- **Singleton** pattern (`factory` constructor)
- **Dio** with 120s connect/receive timeout
- **Interceptor** automatically:
  - Attaches `auth-token` and `Authorization: Bearer` headers from `SecureStorage`
  - Fixes invalid `cycleType: 'pregnancy' → 'regular'` (global enum fix)
  - Logs all requests
- Methods: `get()`, `post()`, `put()`, `delete()`, `postApiMultiPart()`, `putApiMultiPart()`
- Error responses (400, 401, 403, 404, 409, 422) are returned as data, not thrown

## Repository (`Repository`)

- Extends `ChangeNotifier`
- Receives `NetworkApiServices` via constructor
- 65+ methods covering all features (auth, pregnancy, menstrual, postpartum, baby growth, booking, AI chat, subscriptions, policies, etc.)
- Each method calls the appropriate endpoint and parses the response into a typed model

## API Endpoint Groups (from `end_points.dart`):

| Group | Prefix | Examples |
|-------|--------|----------|
|       |        |          |

|                    |                  |                                                                                                                               |
|--------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Auth</b>        | /auths/          | request-verification, verification, set-password, login-with-password, request-forget-password                                |
| <b>Users</b>       | /users/          | getUser, profile-update, onboarding/complete, check-mode-onboard/                                                             |
| <b>Pregnancy</b>   | /pregnancies/    | info, doctor-appointment(s), dashboard/predict, logs                                                                          |
| <b>Menstrual</b>   | /menstruals/     | add-cycle, cycles, logs, predict, dashboard/*                                                                                 |
| <b>Postpartum</b>  | /postpartums/    | add, doctor-appointment(s), recovery-task/add/, feeding, logs                                                                 |
| <b>Baby Growth</b> | /babygrowths/    | add, get, add/milestone, get/milestones, add/photo, get/photos, get/vaccinations, update/vaccination/status/, add/baby-growth |
| <b>Bookings</b>    | /bookings/       | upcoming, past, add, cancel/, doctor/{id}/available-slots                                                                     |
| <b>Doctors</b>     | /doctors/        | doctors                                                                                                                       |
| <b>Medical</b>     | /medical-record/ | upload, gettall                                                                                                               |

|                                |                                           |                                                      |
|--------------------------------|-------------------------------------------|------------------------------------------------------|
| <b>Reco<br/>rds</b>            |                                           |                                                      |
| <b>AI<br/>Chat</b>             | <code>/aichats/</code>                    | <code>create-chat, all-messages, send-message</code> |
| <b>AI<br/>Insig<br/>hts</b>    | <code>/ai/</code>                         | <code>insights</code>                                |
| <b>Com<br/>muni<br/>ties</b>   | <code>/communities/</code>                | <code>all, toggle-like/, add</code>                  |
| <b>Subs<br/>cripti<br/>ons</b> | <code>/plans/,<br/>/subscriptions/</code> | <code>get-all, add</code>                            |
| <b>Agor<br/>a</b>              | <code>/agoras/</code>                     | <code>rtc</code>                                     |
| <b>File<br/>Uplo<br/>ad</b>    | <code>/fileuploads/</code>                | <code>single</code>                                  |
| <b>Polici<br/>es</b>           | <code>/policies/</code>                   | <code>(root), return, terms, shipping</code>         |

## Local Storage

Two storage systems are used:

## 1. `SecureStorage` (`lib/app/data/storage/secure_storage.dart`)

Uses `flutter_secure_storage`. Stores sensitive data:

| Key                     | Methods                                                                                             | Purpose                    |
|-------------------------|-----------------------------------------------------------------------------------------------------|----------------------------|
| <code>token</code>      | <code>saveToken</code> , <code>getToken</code> , <code>clearToken</code>                            | Auth JWT token             |
| <code>user_id</code>    | <code>saveUserId</code> , <code>getUserId</code> , <code>clearUserId</code>                         | Logged-in user ID          |
| <code>tracker_id</code> | <code>saveTrackerId</code> , <code>getTrackerId</code> , <code>clearTrackerId</code>                | Pregnancy/cycle tracker ID |
| <code>con_Id</code>     | <code>saveConversationId</code> , <code>getConversationId</code> , <code>clearConversationId</code> | AI chat conversation ID    |

Has a `clearAll()` method to wipe all keys (used on logout).

## 2. `UserLocalData` (`lib/app/data/storage/user_local_data.dart`)

Also uses `flutter_secure_storage`. Stores app state:

| Key               | Methods                                                               | Purpose         |
|-------------------|-----------------------------------------------------------------------|-----------------|
| <code>role</code> | <code>saveRole</code> , <code>getRole</code> , <code>clearRole</code> | User flow role  |
| <code>step</code> | <code>saveStep</code> , <code>getStep</code> , <code>clearStep</code> | Onboarding step |

|                                  |                                                 |                                     |
|----------------------------------|-------------------------------------------------|-------------------------------------|
| doctor                           | saveDoctorId, getDoctorId                       | Selected doctor ID                  |
| last_stage_screen_shown          | saveLastStageScreenShown, shouldShowStageScreen | Rate-limits stage screen to 30 days |
| post_pregnancy_setup_complete    | save/is/clear                                   | Setup flag                          |
| pregnancy_setup_complete         | save/is/clear                                   | Setup flag                          |
| conception_date                  | save/get                                        | Onboarding data                     |
| baby_name, baby_dob, baby_gender | saveBabyDetails, getBabyDetails                 | Baby info                           |

### 3. UserPreference

(lib/app/services/user\_preference/user\_preference.dart)

Uses **Hive**. Lightweight — only stores `isFirstTime` boolean to control onboarding display.

## Authentication

### Email/Password Flow

1. **Sign Up:** request-verification → OTP → verification → set-password → onboarding/complete
2. **Sign In:** login-with-password → token saved to SecureStorage
3. **Forgot Password:** request-forget-password → OTP verify → set-password

Managed by controllers in `lib/app/controller/sign_in/`, `create_account/`, `forgot_password/`.

## Google Sign-In Flow

**File:** `lib/app/services/social_login/social_login.dart`

- Singleton `SocialLoginService`
- Uses `google_sign_in` + `firebase_auth`
- Reads `SERVERClientID` from `.env` (Web client ID from Google Cloud Console)
- Sign-out before sign-in to always show account picker
- Uses `authenticate(scopeHint: ['email'])` (Credential Manager API)
- On success → Firebase credential exchange → navigates to `stagesView`
- Error handling for API Exception 10, sign-in canceled

## Token Injection

Auth token is automatically attached to every API request via Dio interceptor in `NetworkApiServices`.

---

## Feature Breakdown

### Pre-Pregnancy Flow

**Location:** `lib/app/view/Pre_pregnancy_flow/` + `lib/app/controller/pre_pregenancy_flow/`

| Feature                  | View                                                 | Controller                              |
|--------------------------|------------------------------------------------------|-----------------------------------------|
| Menstrual Cycle Tracking | <code>Mentrua_cycle.dart</code>                      | <code>cycle_celender_controller/</code> |
| Cycle Calendar           | <code>cycle_calendar/cycle_calendar_view.dart</code> | <code>cycle_celender_controller/</code> |



|             |                                   |                                            |
|-------------|-----------------------------------|--------------------------------------------|
| Daily Logs  | Daily_Logs.dart                   | daily_logs_controller/                     |
| AI Insights | aI_insights.dart                  | ai_insights_controller/                    |
| Navigati on | nav_bar/pre_pregnancy_navbar.dart | nav_bar/pre_pregnancy_nav_contr oller.dart |

Pregnancy Flow

Location: lib/app/view/Pregnancy\_Flow/ + lib/app/controller/pregnancy\_flow/

| Feature             | View                        | Controller                |
|---------------------|-----------------------------|---------------------------|
| Pregnancy Home      | pregnancy_home_view.dart    | pregnancy_controller.dart |
| Pregnancy Dashboard | pregnancy_view.dart         | pregnancy_controller.dart |
| Fetal Development   | fetal_development_view.dart | (data from controller)    |
| Appointments        | appointment_view.dart       | pregnancy_controller.dart |
| Add Reminder        | addreminder_view.dart       | (local)                   |

|            |                              |                                         |
|------------|------------------------------|-----------------------------------------|
| Save Log   | savelog_view.dart            | (local)                                 |
| Navigation | navbar/pregnancy/navbar.dart | navbar/pregnancy/navbar_controller.dart |

### Post-Pregnancy Flow

**Location:** lib/app/view/Post\_pregnancy\_flow/ +  
lib/app/controller/post\_pregenancy/

| Feature              | View                           | Controller                      |
|----------------------|--------------------------------|---------------------------------|
| Post-Pregnancy Home  | post_pregnancy_view.dart       | post_pregenancy_controller.dart |
| Baby Details         | baby_detail_view.dart          | (controller)                    |
| Combined Baby Detail | post_pregnancy_pageview.dart   | (pageview)                      |
| Post Appointments    | post_pre_addAppointment.dart   | post_appoitment_controller.dart |
| Baby Growth (Post)   | post_pre_baby_growth_view.dart | baby_growth_controller.dart     |

### Baby Growth

**Location:** lib/app/view/baby\_growth/

| Feature | View File |
|---------|-----------|
|         |           |

|                       |                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------|
| Baby Growth Dashboard | <code>baby_growth_view/baby_growth_view.dart</code>                                 |
| Growth Summary        | <code>baby_growth_view/baby_growth_summary_view.dart</code>                         |
| Milestones            | <code>milestones/mild_stones_view.dart,</code><br><code>add_mild_stones.dart</code> |
| Photos                | <code>photo/photo_view.dart,</code> <code>add_new_photo_view.dart</code>            |
| Vaccinations          | <code>vaccianations/vaccination_view.dart</code>                                    |

## Expert Consultation

**Location:** `lib/app/view/expert_consultation/` +  
`lib/app/controller/experts_consultation/`

| Feature        | View File                                      | Controller                                        |
|----------------|------------------------------------------------|---------------------------------------------------|
| All Doctors    | <code>all_doctor_view.dart</code>              | <code>experts_consultation_controller.dart</code> |
| Doctor Profile | <code>doctor_profile_view.dart</code>          | (local)                                           |
| Slot Selection | <code>select_slot_time_view.dart</code>        | <code>booking/booking_controller.dart</code>      |
| My Bookings    | <code>my_bookings/my_bookings_view.dart</code> | <code>booking/booking_controller.dart</code>      |

|                      |                                                                |                                              |
|----------------------|----------------------------------------------------------------|----------------------------------------------|
| Booking Confirmation | <code>my_bookings/bookings_view.dart</code><br><code>rt</code> | (local)                                      |
| Medical Records      | <code>records/records_view.dart</code>                         | <code>records/records_controller.dart</code> |
| Reports              | <code>report/report_view.dart</code>                           | (local)                                      |

## Video Calling (Agora)

**Location:** `lib/app/agora/`

| File                                    | Purpose                                                                                      |
|-----------------------------------------|----------------------------------------------------------------------------------------------|
| <code>agora.dart</code>                 | <code>TokenGeneratorModel</code> – parses Agora RTC token from API                           |
| <code>video_call_controller.dart</code> | <code>VideoCallProvider</code> – manages Agora engine, join/leave channel, mic/camera toggle |
| <code>video_call_screen.dart</code>     | Video call UI with local/remote video views                                                  |

## AI Assistant

**Location:** `lib/app/view/ai_assistant/` + `lib/app/controller/ai_assistant/`

- Chat-based AI assistant with speech-to-text support
- Backend endpoints: `create-chat`, `all-messages`, `send-message`
- Conversation ID stored in `SecureStorage`

## Subscriptions

**Location:** `lib/app/view/subscription_unlock_plans/`

- Stage-specific subscription screens: Pre-Pregnancy, Pregnancy, Post-Pregnancy, Baby/Child

- Controller:  
`subscription_unlock_plans/controller/subscription_controller.dart`

## Profile

**Location:** `lib/app/view/profile_screen/`

- Profile viewing & editing (`profile_screen.dart`, `profile_update_screen.dart`)
- Sub-screens for policies: Privacy, Terms, Refund, Shipping

## Push Notifications

**File:** `lib/app/push_notification/push_notifications.dart`

- Firebase Cloud Messaging setup
  - Token retrieval happens at boot in `main.dart`
- 

## Theme & Styling

**Location:** `lib/app/theme/`

| File                          | Purpose                                                                 |
|-------------------------------|-------------------------------------------------------------------------|
| <code>app_colors.dart</code>  | Static color constants ( <code>AppColors.backgroundColor</code> , etc.) |
| <code>font_family.dart</code> | Font family name constants (Gilroy family, 10 weights)                  |
| <code>font_style.dart</code>  | Pre-defined <code>TextStyle</code> objects for reuse                    |

**Font Family:** Gilroy (UltraLight → Black, 10 weights registered in `pubspec.yaml`)

## Constants

**Location:** `lib/app/constants/`

| File                     | Purpose                                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------------------------------|
| <code>flow.dart</code>   | <code>FlowType</code> enum: <code>postPregnancy</code> , <code>pregnancy</code> , <code>prePregnancy</code> |
| <code>images.dart</code> | Static asset path strings for all image files                                                               |

## Reusable Widgets

Location: `lib/app/widgets/` (28 files)

| Widget          | File                                    | Purpose                            |
|-----------------|-----------------------------------------|------------------------------------|
| Button          | <code>button.dart</code>                | Standard app button                |
| Custom Image    | <code>custom_image.dart</code>          | Cached network image with fallback |
| Custom AppBar   | <code>custom_appbar.dart</code>         | Consistent top navigation bar      |
| Text Form Field | <code>custom_textform_field.dart</code> | Styled text input                  |

|                  |                                                |                                               |
|------------------|------------------------------------------------|-----------------------------------------------|
| Date<br>Picker   | <code>common_select_date_textfield.dart</code> | Date<br>selection field                       |
| Timer<br>Field   | <code>CustomTimerFormField.dart</code>         | Time input                                    |
| Days<br>Picker   | <code>custom_days_picker.dart</code>           | Multi-day<br>selection                        |
| Loading          | <code>custom_loading.dart</code>               | Loading<br>spinner<br>overlay                 |
| No Data<br>Found | <code>custom_no_data_found.dart</code>         | Empty state<br>display                        |
| Popup            | <code>app_popup.dart</code>                    | Toast/dialog<br>helper                        |
| Pinput           | <code>pinput.dart</code>                       | OTP input<br>field styling                    |
| Validatio<br>n   | <code>validation.dart</code>                   | Form<br>validators<br>(email,<br>phone, etc.) |
| Switch<br>Button | <code>switch_btn.dart</code>                   | Toggle switch                                 |

|                       |                                                                                                         |                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------|
| Gradient<br>Checkbox  | <code>gradient_checkbox.dart</code>                                                                     | Styled<br>checkbox                |
| Progress<br>Bar       | <code>gradientprogressBar.dart</code>                                                                   | Gradient<br>progress<br>indicator |
| Circular<br>Indicator | <code>circular_indicator.dart</code>                                                                    | Circular<br>progress              |
| Slider                | <code>slider_widget.dart</code>                                                                         | Custom slider                     |
| Container<br>s        | <code>container.dart</code> , <code>custom_cont.dart</code> ,<br><code>sub_custom_container.dart</code> | Styled<br>wrappers                |
| SizedBox<br>helpers   | <code>sizedbox.dart</code>                                                                              | Common<br>spacing                 |
| Text<br>helper        | <code>text.dart</code>                                                                                  | Text widget<br>shortcuts          |
| TextField<br>title    | <code>textfield_title.dart</code>                                                                       | Label above<br>fields             |
| Icon +<br>Text Row    | <code>icon_text_row.dart</code>                                                                         | Horizontal<br>icon-text<br>layout |



|                   |                                                                                      |                                                                                      |
|-------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Print utility     | <code>print.dart</code>                                                              | <code>pt()</code> — custom logger <b>(always use this, not <code>print()</code>)</b> |
| General Exception | <code>general_exception.dart</code>                                                  | Error screen UI                                                                      |
| Feeding Dialogs   | <code>feeding_details_dialog.dart</code> ,<br><code>feeding_entry_dialog.dart</code> | Feeding log popups                                                                   |

## Environment Configuration

### `.env` file (`assets/.env`)

`SERVERClientID=<Google Web Client ID>`

- Loaded at app boot via `flutter_dotenv`
- Used by `SocialLoginService` for Google Sign-In

## Firebase

- Config in `lib/firebase_options.dart` (auto-generated by FlutterFire CLI)
- `firebase.json` at project root for hosting config

## Android Setup

- `android/` directory contains standard Flutter Android project
- SHA-1 must be configured in Firebase Console for Google Sign-In

## Existing Documentation

| File                                   | Contents                                 |
|----------------------------------------|------------------------------------------|
| <code>docs/API_DOCUMENTATION.md</code> | Full API endpoint documentation          |
| <code>docs/CURL_EXAMPLES.md</code>     | cURL commands for testing every endpoint |
| <code>docs/swagger.json</code>         | OpenAPI 3.0 specification                |
| <code>structure.md</code>              | Brief architecture overview              |
| <code>gemini.md</code>                 | AI assistant coding constraints          |

---

## Code Conventions

| Area                   | Convention                                                                                                                  |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| File naming            | <code>snake_case</code> (e.g., <code>splash_view.dart</code> , <code>booking_controller.dart</code> )                       |
| Class naming           | <code>PascalCase</code> (e.g., <code>SplashView</code> , <code>BookingController</code> )                                   |
| Variable/Method naming | <code>camelCase</code> (e.g., <code>currentUser</code> , <code>fetchData()</code> )                                         |
| Screens                | <code>StatefulWidget</code> with <code>Scaffold</code> + <code>SafeArea</code> , or <code>StatelessWidget</code> for static |

|                     |                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Logging</b>      | Use <code>pt("message")</code> from <code>lib/app/widgets/print.dart</code> . <b>Never use <code>print()</code></b>  |
| <b>State access</b> | <code>Consumer&lt;T&gt;</code> for reactivity, <code>Provider.of&lt;T&gt;(context, listen: false)</code> for actions |
| <b>Navigation</b>   | <code>Navigator.pushNamed(context, AppRoutes.routeName)</code>                                                       |

## Known Gotchas & Notes

1. **cycleType Interceptor:** The Dio interceptor globally intercepts and fixes `cycleType: 'pregnancy'` → `'regular'` in all outgoing requests. This was a workaround for a backend enum validation issue.
2. **Backend is separate.** The backend code is at `C:\Users\telig\Downloads\Compressed\Baby-Land-Node-Server-main`. **Never modify backend code.**
3. **BookingController is registered twice** in the MultiProvider list in `main.dart`. This is an existing quirk — be aware of potential conflicts.
4. **File naming inconsistencies:** Some view directories use `PascalCase` (e.g., `Post_pregnancy_flow/`, `Pre_pregnancy_flow/`, `Pregnancy_Flow/`) while controllers use `snake_case`. This is inconsistent but should not be changed.
5. **Spelling variations:** Some names have spelling variations (e.g., `pregenancy` vs `pregnancy`, `mentural` vs `menstrual`, `mild_stones` vs `milestones`, `celender` vs `calendar`). These are embedded in file names and class names — do not rename.
6. **Common Profile Header:** The `GetUserProvider` and `ProfileHeader` widget in `lib/app/common_profile_header/` are used across multiple screens for showing the user's profile info.
7. **`UserLocalData.shouldShowStageScreen()`:** Controls whether to show the stage selection screen — it's rate-limited to once every 30 days.

## How to Run

# 1. Clone the repository

```
git clone <repo-url>
```

# 2. Install dependencies

```
flutter pub get
```

# 3. Set up environment

# Ensure assets/.env has the correct SERVERClientID

# 4. Firebase setup

# Ensure firebase\_options.dart is configured for your Firebase project

# Add your SHA-1 to Firebase Console for Google Sign-In

# 5. Run the app

```
flutter run
```

# 6. For Android release

```
flutter build apk
```

## Prerequisites

- Flutter SDK ^3.9.2
- Android Studio / Xcode
- Firebase project configured
- Google Cloud Console project with OAuth 2.0 Web Client ID (for Google Sign-In)
- Agora App ID (for video calling — configured on backend)

